

CODEMIX SKILL

The safety + decision layer between AI agents and their actions

We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.

Initiative: Codemix Skill • Owner: Team Ramanathan & Sadhana • Track 1: Customer & Employee Experience

● THE GAP

Voice agents can immediately call a refund, cancellation or replacement API. One missed negation turns the right conversation into the wrong transaction.

● OUR PRODUCT

A plug-in between the AI agent and its API/tool calls. It validates intent, negation, confidence, sentiment and SOP rules before execution.

● FRESHWORKS EXTENSION

Unsafe, uncertain or escalated calls become Freshdesk tickets with transcript, decision context and action details for a human agent.

Understand

Decide

Act

Audit

*“Order cancel **mat** karo, bas address change karna hai.”*

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Problem Statement

In code-mixed voice, one short word can reverse the meaning of an API action

HINDI EXAMPLE

*“Order cancel **mat** karo, bas address change karna hai.”*

Correct intent: do NOT cancel → update address

TAMIL EXAMPLE

*“Refund **vendam**, replacement anuppunga.”*

Correct intent: do NOT refund → send replacement

HOW IT BREAKS TODAY

Caller says “don't cancel”

Speech-to-text drops
“mat”, or the intent model
ignores it

Agent calls cancel_order()

**Order cancelled: money,
reshipping, trust lost**

48%

of negated requests would still trigger a
refund / cancel in our own Round-1
engine

*Measured: 10 of 21 team-written test sentences
(Hindi, Tamil, English, Kannada)*

26%

of cash-on-delivery orders in India return
to origin, so refund / cancel / return calls
are high-volume

Source: Shipway ShipNotes report, FY25 (Jul 2025)

Why Now & Competitive Landscape

AI agents have write access, and nobody gates what they are allowed to do on a misheard word

WHY NOW



AI agents can act

Tool calling and MCP let agents trigger refunds, cancellations and returns directly.



Code-mixing is normal

Indian callers switch language mid-sentence; the risk sits in small intent-flipping words.



The voice stack is ready

VoBiz, Sarvam, Claude and ElevenLabs make a checked, real-time voice agent possible now.

27-70% **Hinglish word error rate across speech-to-text models.** Speech recognition alone can't be trusted with irreversible actions. *Deepgram*

Our timing Not another voice bot: the layer that decides when an agent may call an API.

COMPETITIVE LANDSCAPE

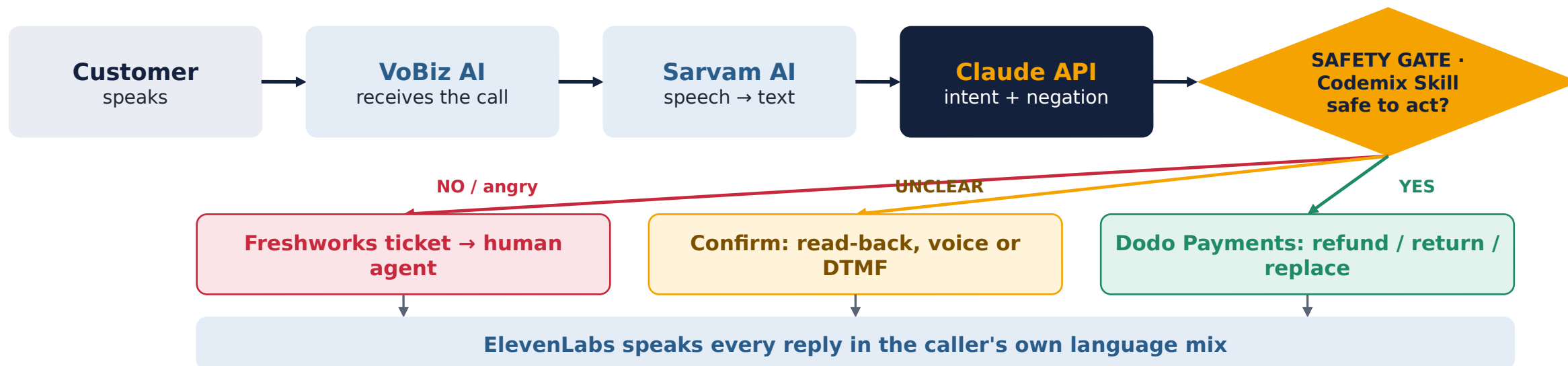
Alternative	Strength	Gap
Do nothing	Zero effort	Risky actions run on the first interpretation
Confirm-everything IVR	Safe confirmation	Friction on every action, even low-risk ones
Generic LLM guardrails	Policy, toxicity, PII	Not an action gate for code-mixed negation + speech confidence
Codemix Skill	Code-mixed intent → risk check → ALLOW / CONFIRM / BLOCK	Unsafe cases go to Freshworks with full context

Our edge Risk-based action gate + human escalation + a full audit record.

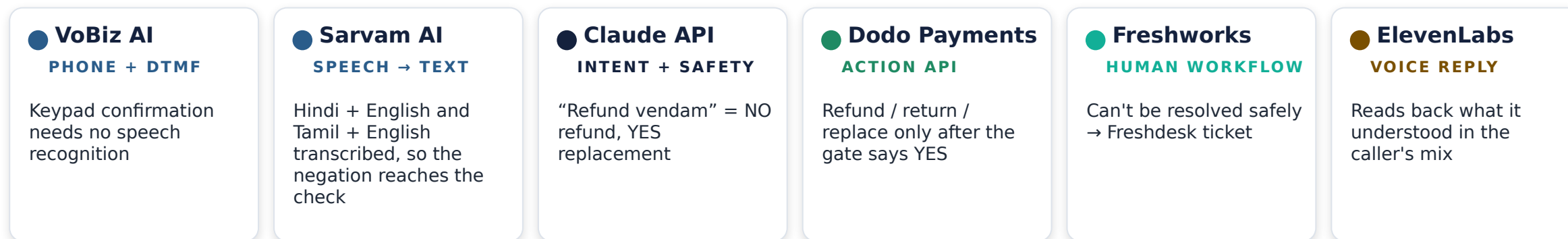
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Solution: the Intent-Validation Safety Gate

Between what the voice agent hears and what it is allowed to do; each part has one job in preventing a wrong action

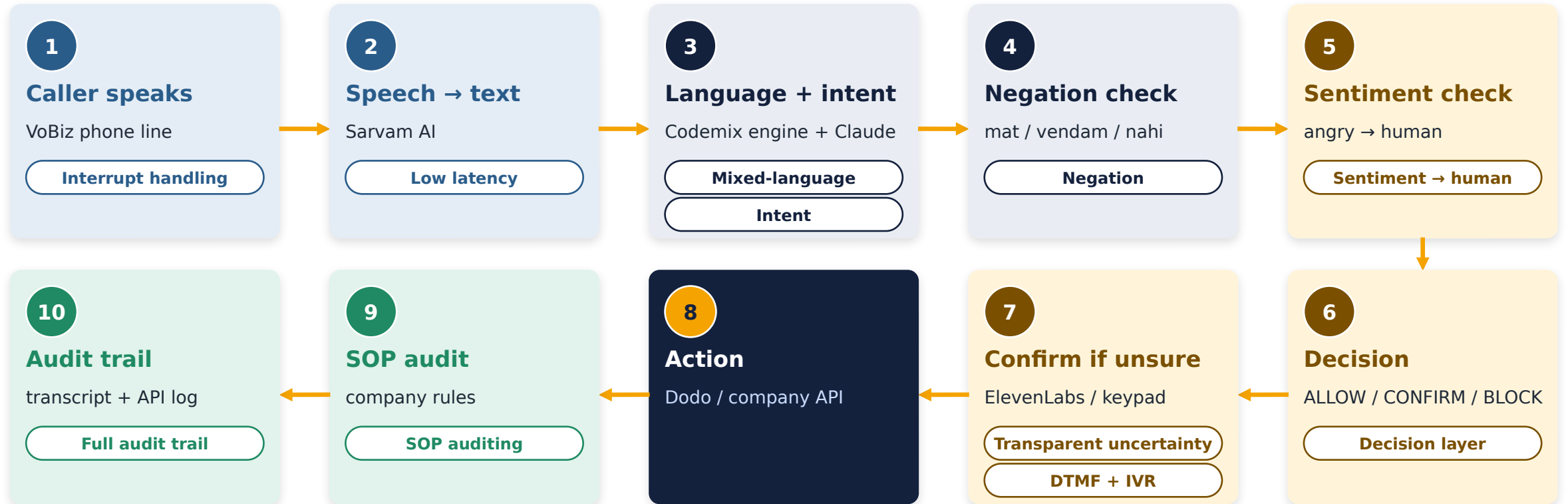


HOW EACH PART PREVENTS A WRONG ACTION



5 One Call, End to End — All 11 Features

Every feature appears only where it adds value; routine calls take the fast path



Sentiment: Claude scores caller sentiment every turn; two negative turns or an explicit request for a human → Freshdesk ticket.

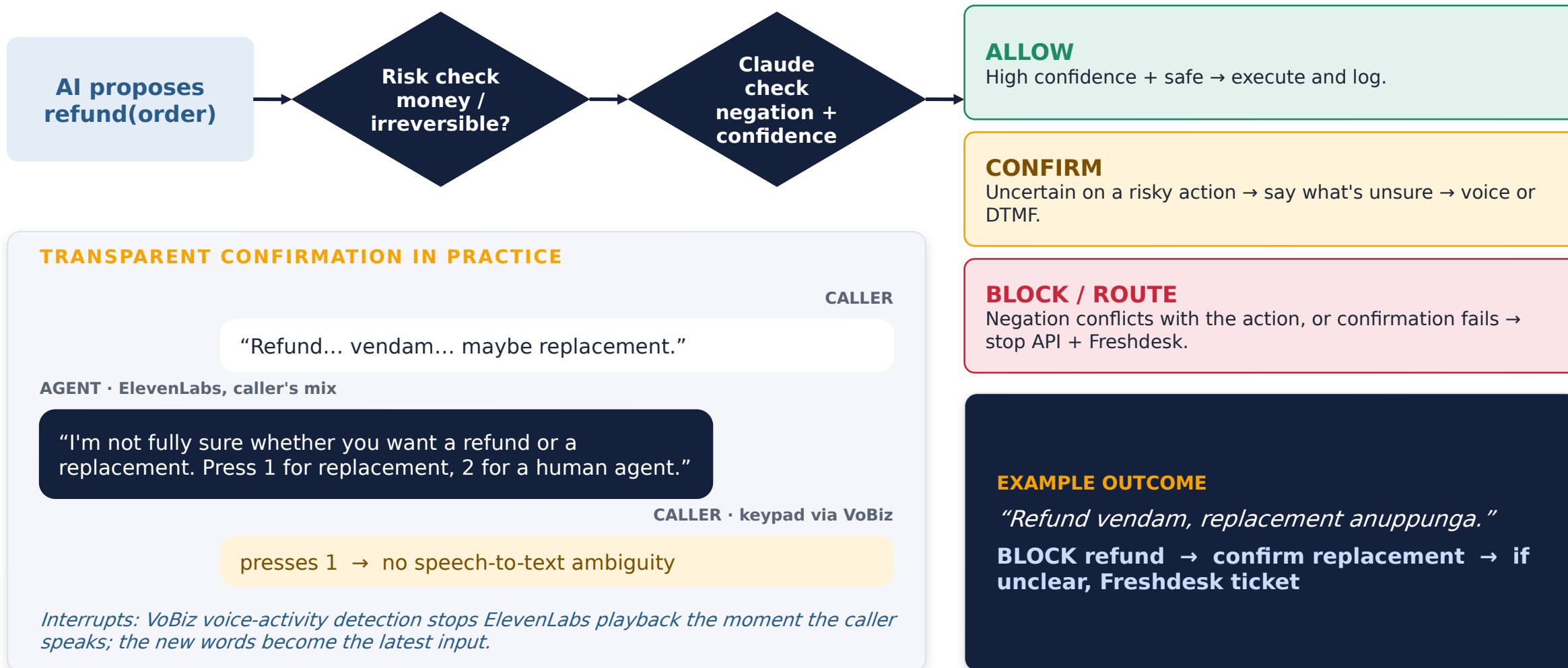
Low latency: fast offline checks first (1.87 ms measured); Claude runs only before sensitive actions.

SOP audit: every call is checked against company rules (authorisation, required confirmation, escalation) → PASS / VIOLATION + audit reference; transcript, decisions and API calls are stored.

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The Safety Decision & Transparent Confirmation

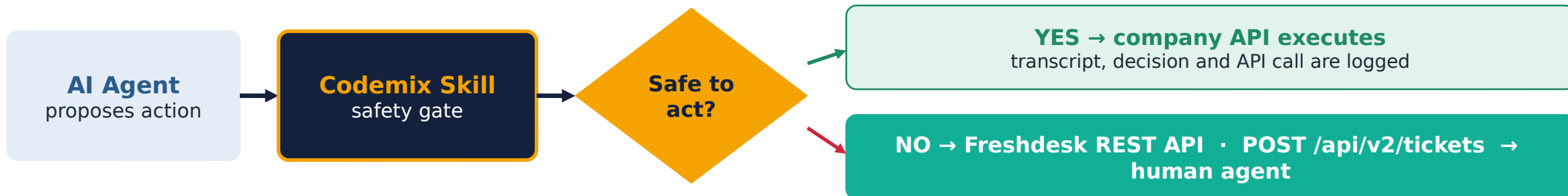
The agent never silently guesses on an irreversible action



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Freshworks Integration — Human-in-the-Loop

Freshdesk takes the handoff; also works with Freshcaller as the voice channel and alongside Freddy AI Agent



A ticket is created when...

- the action is unsafe or unclear
- the caller doesn't confirm
- the caller is angry / distressed
- the AI should not guess



What we add to Freshdesk

Code-mixed intent validation, action gating, transparent confirmation and structured escalation, on top of ticketing, agent workspace and customer history.

What the human agent receives in the Freshdesk ticket

CONVERSATION

Call ID + timestamp

Original code-mixed transcript

Detected languages

English summary

Voice / DTMF source

DECISION

Intent + confidence

Negation detected

Safety verdict

Requested API action

Confirmation status

OPERATIONAL

Priority / tags

SOP result

Reason for escalation

Attempted API call

Audit reference

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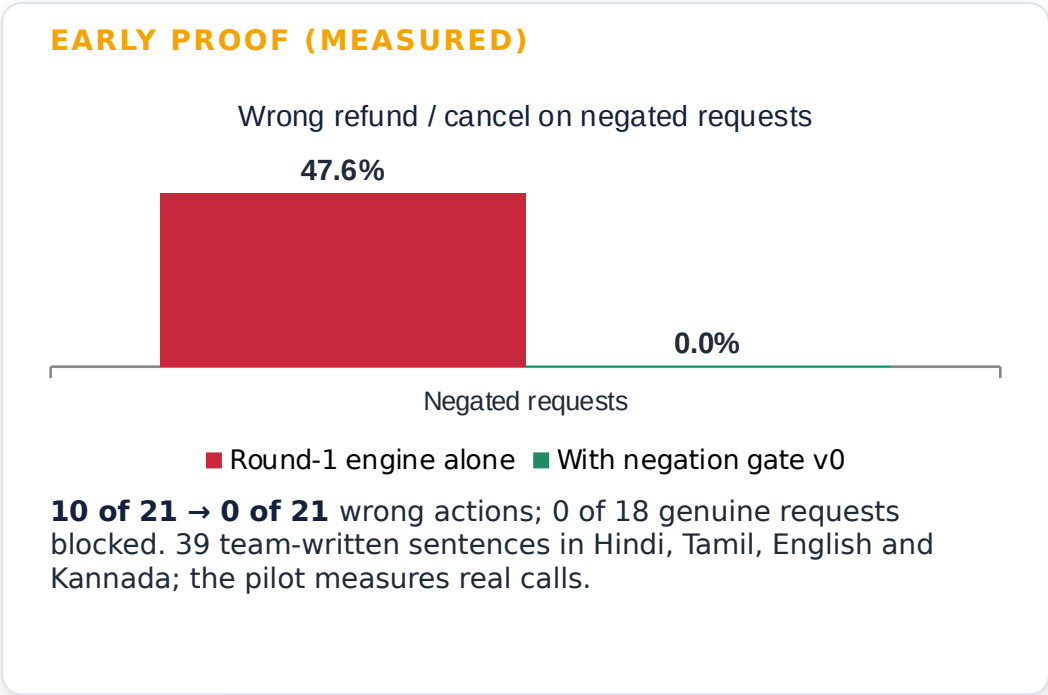
Agentic Design, MVP Scope & Early Proof

A plug-in control point that intercepts the proposed action before the company API is called

AGENTIC DESIGN: THE PLUG-IN CONTROL POINT



BUILT TODAY	Code-mix engine + intent scoring · MCP server · live Freshdesk ticket API · negation gate v0 + benchmark · Vercel demo
MVP (3 MONTHS)	VoBiz + Sarvam + ElevenLabs voice path · Claude safety gate · Dodo action path · Freshdesk escalation · Hindi + Tamil + English + Kannada · audit trail
NEXT	Sentiment + interrupt tuning · SOP rule editor + reports · more Indian languages · Freshworks Marketplace app



Customers, Value, Pricing & Go-to-Market

Start where AI agents already have permission to act; pilot → proof → Freshworks distribution

6 Target Customers

Primary: D2C, e-commerce and fintech brands on Freshdesk deploying AI voice agents for Indian customers.

Secondary: BPOs needing one safety / policy layer across many clients.

Protected actions: refunds, returns, replacements, cancellations.

Why now: ~26% of COD orders return to origin (Shipway FY25), so these calls are high-volume.

7 Value Proposition

Plug it in once: the existing agent serves mixed-language callers, and nothing irreversible happens on a misheard word.

Without

- Acts on first guess
- Dropped “mat” = wrong refund
- Blind escalation, no record

With Codemix Skill

- Checks intent + negation first
- Confirms when unsure
- Freshdesk handoff + audit trail

8 Pricing & Packaging (proposed)

Free

Monitor mode: flags risky actions, never blocks

\$0.03

per guarded action: plug-in on the customer's own agent

₹8 / min

full voice agent: VoBiz + Sarvam + Claude + ElevenLabs + Freshdesk

Custom

Enterprise: SOP rule packs, India data residency, SLA

Priced above run cost (~55% gross margin, slide 10). One prevented ₹1,000 wrong refund pays for ~390 guarded actions.

9 Go-to-Market Plan

- 1 3 Indian D2C design partners on Freshdesk
- 2 Free monitor-mode pilot
- 3 Publish wrong-action proof + demo video
- 4 Freshworks Marketplace listing
- 5 Scale via Freshdesk deals + BPO partners

Costs, Savings & Revenue, Metrics & Risks

What it costs, who gains what, how we measure it, and what could go wrong

10 Costs & Unit Economics

~\$90K

3-month MVP build: M1 data · M2 gate + voice · M3 Freshdesk + pilot

Engineers \$60K · linguist \$10K · PM \$8K · APIs \$7K · buffer \$5K

RUN COST PER 3-MIN VOICE CALL

VoBiz telephony (₹0.45/min)	₹1.35
Sarvam STT (₹30/hr)	₹1.50
ElevenLabs Flash (\$0.05/1K chars)	₹6.40
Claude checks (Haiku 4.5 / Sonnet 5)	₹0.71
Hosting + transcript storage	₹0.43
Total ≈ ₹3.5/min vs ₹8 price	₹10.4

Plug-in only: ₹1.1 cost vs ₹2.55 price per action. Dodo and Freshdesk fees stay on the customer's own plans.

11 Success Metrics

	TODAY	TARGET
Wrong actions on negation	48%*	<0.5%
Engine latency	1.87ms†	<5ms
p95 reply (sensitive)	—	<900ms
Confirmation rate	—	≤15%
Audit coverage	—	100%
Reversal tickets	—	−30%
Adoption (2 qtrs)	—	15%

** Round-1 engine, no gate, 21-sentence test set † npm test average*

12 Risks & Assumptions

STT drops the negation

→ confirm every risky action

Safety check adds latency

→ deep checks only when sensitive

Dialect / spelling variation

→ lexicon + model fallback

Read-back friction

→ ≤15% confirmation budget

ASSUMPTIONS

- Pilot brands expose refund / return APIs
- Callers accept a short confirmation
- Sarvam is accurate enough with the fallback

WHO GAINS WHAT IN YEAR 1

illustrative · 48% misfire rate measured, volumes assumed · details in speaker notes

BRANDS SAVE

≈ ₹6 Cr

wrong refunds & cancellations avoided · ₹3.8 saved for every ₹1 paid

FRESHWORKS EARNS

\$349K

Plug-in \$180K + Voice agent \$169K

GROSS PROFIT

~\$195K

after \$154K run costs · pays back the \$90K build in ~8 months